

Light based Sensors

Camera Sensor (passive)

Light from an object enters through the camera lens and CCD sensor converts the light into electrical signals. This analog signal is then converted to digital values using ADC. Then the camera processes the digital data and saves it as a digital image.

Active IR Sensor

An active IR sensor has an IR LED that sends infrared light and a photodiode that receives the reflected light. When the light returns, the photodiode's voltage changes, which helps detect objects. Some sensors also use pyroelectric material that produces a small voltage when temperature changes.

Passive IR Sensor

A PIR sensor uses a Fresnel lens and pyroelectric materials to detect heat. The lens focuses infrared radiation onto the sensor, which has two halves that normally see equal heat. When a warm object moves, it creates a difference between the halves, and the sensor gives a HIGH output to show motion.

Infrared camera/Thermal camera(passive)

All objects emit infrared heat, and the camera collects it through a lens. The heat falls on a sensor array where each pixel turns it into electrical signals. A processor converts these into temperature values and shows them as colors, creating a thermal image that can be displayed and saved

Infrared Temperature Sensor(passive)

All objects emit infrared heat and the sensor detects the emitted heat using an IR detector(thermopile/pyroelectric material). Detected energy then converted to electrical signal to temperature values.

Infrared Remote (active)

A remote uses an IR LED to send encoded infrared light signals when you press a button. The IR receiver (photodiode) detects these signals and converts them into electrical signals. The system then decodes them and performs the action like changing channel or volume.

UV Sensor (passive)

A UV sensor absorbs ultraviolet light and creates electrons, which changes its electrical signal. More UV light produces a higher voltage or current, while less UV gives a lower output. The circuit processes this signal and shows it as a UV level or intensity.

Pulse Oximeter Sensor (active)

A pulse oximeter uses red and infrared light to pass through the finger, where blood absorbs the light differently based on oxygen level. Oxygenated blood absorbs infrared light more and di-oxygenated blood absorbs red light more. A sensor measures this and calculates oxygen saturation (SpO_2) and heart rate.

Laser Sensor(active)

A laser sensor sends a laser beam to an object and receives the reflected light back through a lens. The position of the reflected light spot on the detector changes

when the object moves. By measuring this shift, the sensor calculates the exact distance or position.

Magnetic and Electromagnetic Sensors

Reed Switch(passive)

A reed switch has two thin metal strips sealed inside a glass tube. Normally, the contacts stay open, so the circuit is OFF. When a magnet comes near, the strips attract each other and close the circuit, turning it ON. Once the magnet is removed, they separate again and turn OFF. Outputs simple ON/OFF switching signal.

Hall Effect Sensor(active)

A Hall effect sensor works when a current-carrying semiconductor material is placed in a magnetic field, creating a small voltage called Hall voltage. The strength of this voltage changes with the strength of the magnetic field. This change is used to produce an output voltage signal that represents the magnetic field level.

Electromagnetic Sensor(active)

This sensor works using electromagnetic induction by creating an alternating magnetic field from a coil. When a metal object comes near, it generates eddy currents that disturb this field. The sensor detects this change and converts it into an electrical signal that shows whether the metal is present and how close it is.

Sound Based Sensors

Microphone(passive)

Sound waves make the microphone diaphragm vibrate when they hit it. These vibrations stress a piezoelectric crystal, creating a small electrical charge due to changes inside the material. The result is a weak electrical signal that matches the sound, which is then amplified and processed into audio.

Ultrasonic Sensor(active)

An ultrasonic sensor sends high-frequency sound waves toward an object using a transmitter. The waves reflect back as an echo, which is detected by the receiver. It measures the time between sending and receiving the echo and uses it to calculate distance.

Temperature Sensors

Resistance Temperature Detector (RTD) (passive)

An RTD measures temperature based on how a metal's resistance changes with heat. When temperature rises, the resistance increases, and when it falls, the resistance decreases. This change is measured and used to find the exact temperature.

Thermocouple(passive)

A thermocouple works when two different metal junctions are at different temperatures, producing a small voltage due to the thermoelectric effect. This voltage changes based on the temperature difference. The device measures this voltage and converts it into temperature.

Thermistor(passive)

A thermistor changes its resistance when temperature changes. The circuit measures this resistance and uses its known relationship with temperature to calculate the value. This allows accurate temperature sensing based on resistance change.

Proximity Sensor

A proximity sensor uses an IR LED that sends infrared light and a detector that receives the reflected light. When an object comes close, more IR light is reflected back to the sensor. The change in reflection tells the device how near the object is and helps detect presence. Used to turn OFF display and disable touch during calls.

Inductive Proximity Sensor

An inductive proximity sensor creates a high-frequency electric field using a coil or electrode. When a metal or object comes close, it changes the field and affects the oscillator's signal. The circuit detects this change and gives an ON/OFF output to show the object's presence.

Capacitive Sensor

A capacitive proximity sensor works by detecting changes in capacitance. The sensor creates an electric field, and when an object comes near, the capacitance of the system increases. This change affects the circuit, and when it crosses a limit, the sensor gives an output signal to show presence.

Microwave Motion Sensor

A microwave (radar) motion sensor continuously sends out microwave signals and listens for the reflected waves. When nothing moves, the reflected signal stays

stable. If an object moves, it changes the reflection pattern, and the sensor detects this change. It then triggers an output like motion detection or an alarm.

Smoke Sensor

An ionization smoke detector uses a small radioactive source to ionize the air between two charged plates, creating a small electric current. When smoke enters, it attaches to the ions and reduces the current flow. The circuit detects this drop and triggers the alarm.

Photoelectric Smoke Sensor

A photoelectric smoke detector uses a light source and a sensor placed at an angle inside a chamber. Normally, the sensor does not receive light, so nothing happens. When smoke enters, it scatters the light toward the sensor, which detects the change and triggers the alarm.

Gas Sensor

A gas sensor usually uses Tin Dioxide (SnO_2), a semiconductor that changes conductivity based on gas presence. In clean air, oxygen traps electrons and keeps conductivity low, but when gas is present, it reacts and releases electrons, increasing conductivity. The sensor detects this change and gives an output showing gas level or presence.

Humidity Sensor

A humidity sensor measures moisture in the air using a material between two electrodes. When humidity changes, its resistance or conductivity changes, and the circuit converts this into a readable signal.

A temperature sensor often uses an NTC thermistor, where resistance decreases as temperature increases and increases as temperature drops. The IC processes both signals and gives digital values for temperature and humidity.

Touch Sensor

Capacitive Touch Sensor

A capacitive touch sensor constantly checks changes in capacitance on its surface. When a finger touches it, the electric field changes and the capacitance increases. The circuit detects this change and sends a signal to show a touch event.

Resistive Touch Sensor

A resistive touch sensor has two conductive layers separated by a small gap. When no touch is applied, there is no contact between them. When pressure is applied, the layers touch, causing a change in voltage that is detected as a touch signal.

Accelerometer

Piezeoelectric Type

An accelerometer measures motion and tilt by sensing acceleration, including gravity. In a piezoelectric type, internal crystals produce a small voltage when they are pressed or stressed by movement. This voltage is proportional to the acceleration and is used to detect motion, direction, and orientation.

Capacitive Type

In a capacitive accelerometer, tiny internal structures act like capacitor plates. When the device moves, the position of a small mass shifts and changes the

distance between the plates. This changes the capacitance, which is then converted into a voltage signal to measure motion.

Gyroscope Sensor

A gyroscope measures how fast something is rotating using the Coriolis force. Inside it, a small structure vibrates in a fixed direction, and when the device rotates, this vibration shifts. The change creates a voltage signal that shows the angular speed and direction of rotation.

Door/Elevator Sensor

An IR sensor pair (TX and RX) is used as a safety system for automatic doors. The transmitter sends an infrared beam to the receiver, and the door works normally when the beam is continuous. If someone blocks the beam, the receiver loses the signal, and the door stops or reopens for safety.

Fingerprint Sensor

Optical Fingerprint Sensor

An optical fingerprint sensor shines light on the finger and captures the reflected image using a camera or sensor. The image is processed to detect ridge and valley patterns. These patterns are then converted into digital binary data for identification.

Capacitive Fingerprint Sensor

A capacitive fingerprint sensor uses a grid of tiny capacitors and the natural conductivity of your finger. When you place your finger on it, the ridges change

the capacitance while the valleys do not. The circuit reads this pattern and converts it into a digital fingerprint for authentication.

LiDAR Sensor

A LiDAR sensor sends fast laser pulses toward an object and measures how long they take to come back after reflection. Using this time and the speed of light, it calculates distance (the return time is divided by 2 because the light travels to the object and back). This allows the sensor to create accurate distance measurements and even build a 3D surface map.

Water Level Sensor

A capacitive water level sensor measures liquid level by detecting changes in capacitance. The tank wall and sensor act like a capacitor, and the liquid changes the dielectric value. As the level rises, capacitance increases, and the circuit converts this into a signal showing how full the tank is.